ml^{-1}) is added to the tank, and 10 ml withdrawn after 2 h of settling; measurements yield a net value (background subtracted) of 500 counts in 10 min. Calculate the free liquid volume in the tank.

18.11. Calculate the critical deposition potential ($E - E^0$) for 10^{-12} M ^{210}Bi on a gold cathode (no over-voltage) from the Nernst equation (18.6), where the chemical activity of the reduced state (Bi^0) is set to unity.

18.12. A mineral ore contains cobalt and small amounts of nickel. In order to determine the nickel concentration it must be separated from cobalt. Solvent extraction using 0.01 M 8-hydroxyquinoline in $CHCl_3$ is chosen. Which metal should be extracted from the other, and at what pH? Consider Figure 18.3 and connected text.

18.13. In a solvent extraction system consisting of uranium and lanthanum in 1 M HNO_3 and 100% TBP, $D_U = 20$ and $D_{La} = 0.07$. If a phase ratio $\theta = V_{org} / V_{aq} = 0.5$ is chosen, how much uranium is removed from the aqueous phase in three repeated extractions? How much of the lanthanum is co-extracted? The fraction extracted with n fresh organic volumes (V_{org}) from one aqueous volume (V_{aq}) is:

$$E_n = 1 - (1 + D\theta)^{-n} \tag{18.40}$$

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CHAPTER 19

Principles of Nuclear Power

Contents

19.1. The Nuclear Reactor 597
19.2. Energy Release in Fission 601
19.3. Fission Probability 602
19.4. The Fission Factor 605
19.5. Neutron Moderation 606
19.6. The Neutron Cycle 607
19.7. Neutron Leakage and Critical Size 611
19.8. Reactor Kinetics 613
19.9. Fuel Utilization 616
19.10. The Oklo Phenomenon 621
19.11. Reactor Concepts 623
19.12. Research and Test Reactors 624
19.13. Thermal Power Reactors 625
19.13.1. The pressurized water reactor 625
19.13.2. The boiling water reactor 626
19.14. Power Station Efficiency 630
19.15. Reactor Safety 632
19.16. Radioactive Reactor Waste 638
19.17. Nuclear Fusion 639
19.17.1. The controlled thermonuclear reactor 639
19.17.1.1. Inertial confinement 640
19.17.1.2. Magnetic confinement 641
19.17.2. Experimental fusion reactors 641
19.17.3. Technical issues of large fusion reactors 642
19.18. Nuclear Explosives 646
19.19. Exercises 651
19.20. Literature 653

From earliest times man has become increasingly dependent of a variety of energy sources: heat from burning of wood, animal dung, coal, oil, natural gas, etc., mechanical energy from steam engines, wind mills and water falls, and electricity from the same sources as well as from nuclear reactors. Much effort has been invested in searches for new energy sources, preferably of the "renewable" kind in contrast to the non-renewable sources[1]. Energy is a strategic resource, playing an important role in international and

[1] In a wider context, all energy originates from gravitational collapse, annihilation, fusion reactions or mass loss into black holes and its sources are thus never renewable.

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